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Fleet resupply by drones for last-mile delivery

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ABSTRACT

The *Vehicle Routing Problem with Release Dates and Drone Resupply* consists of routing a fleet of trucks to deliver orders that arrive at a depot over time. During the delivery horizon, the trucks can return to the depot to collect newly arrived orders, or these orders can be resupplied to the trucks along their routes via drones dispatched from the depot. A Mixed-Integer Linear Programming (MILP) formulation is developed for the version of the problem where order arrival times at the depot (generally termed order release dates) are known beforehand. To address large-size instances, we devise a unified matheuristic approach that provides high-quality solutions for both the truck-and-drone and the truck-only versions of the problem. In this approach, truck routes are iteratively modified using a tabu search scheme, where a subordinate fast MILP model defines optimal loading operations (truck depot returns and drone resupplies) for promising truck routes. We perform extensive numerical experiments with instances of up to 100 customers. Results show the effectiveness of the matheuristic approach for solving both the truck-and-drone and the truck-only versions of the problem. We also show the benefits of drone resupply to reduce completion times and the number of times the trucks need to return to the depot to collect newly released orders. Furthermore, we provide several managerial insights regarding fleet utilization and consolidation.

1. Introduction

The interest in optimizing last-mile logistics using drones has grown considerably in recent years (Chung et al., 2020; Macrina et al., 2020). Compared to parcel delivery using conventional vehicles only, drones represent a potentially faster and cheaper alternative, with less environmental impact (Raconteur, 2018; Stolaroff et al., 2018).

The first to study the use of drones to reduce last-mile delivery costs were Murray and Chu (2015), who proposed an innovative delivery system where a drone is periodically launched en route from the dispatch vehicle for autonomous delivery to the customer. Since then, most research efforts have focused on using drones to make direct deliveries to consumers (see Moshref-Javadi & Winkenbach, 2021 and Dukkanci et al., 2023 for recent surveys on drone logistics). Nevertheless, the physical process of how a drone would hand off a parcel to a customer has yet to be resolved, especially in urban areas as protocols and regulations do not yet exist for most last-mile drone applications (Boysen et al., 2021; Cornell et al., 2023; Gaba & Winkenbach, 2020).

In this paper, we investigate the use of drones for sending newly available orders to delivery trucks while the trucks are en route. This approach enables last-mile logistics operators to plan deliveries to customers whose orders are not available when the trucks first leave the warehouse, but are expected to become available (at specific times) later in the day and should be delivered that same day. These newly available orders can be transferred to the trucks along their routes via drones dispatched from the warehouse. Compared to drone delivery directly to customers, this approach would be significantly easier to implement in practice, as trained delivery drivers, instead of untrained customers, would be the only ones to receive the drones, unload orders, and send the drones back to the warehouse.

Considering a fleet of trucks and drones, we formally introduce the *Vehicle Routing Problem with Release Dates and Drone Resupply* (VRPRD-DR), which consists of routing a fleet of uncapacitated trucks to deliver orders that become available at a depot over time. The VRPRD-DR considers two loading strategies to manage the arrival of new orders: trucks can return to the depot to collect them, or these new orders can be resupplied to the trucks at particular customer locations via drones. The aim of the VRPRD-DR is to find the routing and loading decisions such that the completion time (i.e., the makespan) of the delivery process is minimized, which is when the last truck returns to the depot from its final tour.

We develop a Mixed-Integer Linear Programming (MILP) formulation for the version of the problem where customer locations and order ready times at the depot (most often termed order release dates)

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are known beforehand. Applications of this problem arise primarily in urban logistics where last-mile distributors have orders arriving at their local facilities (from the previous distribution leg) at known times during the day for same-day delivery (Cattaruzza et al., 2016a). Other use cases are found in make-to-order operations, such as pharmacies or perishable goods, where planners know when each order will be prepared and ready to be dispatched (Shelbourne et al., 2017).

Due to the inherent complexity of the VRPRD-DR, we then develop a matheuristic approach for solving large-size instances, which is based on a tabu search scheme. In this approach, we iteratively modify the truck routes and define optimal loading operations (i.e., drone resupplies and truck depot returns) through a fast subordinate MILP. To speed up the matheuristic, we introduce two pruning strategies to run the subordinate MILP only for promising truck routes. An appealing property of the matheuristic is that it provides high-quality solutions for both the truck-and-drone and the truck-only versions of the problem.

Note that the proposed MILP formulation and matheuristic approach are not specific to trucks and drones, but are readily applicable to other vehicle combinations, such as a delivery van combined with a cargo bike for resupply. Nevertheless, drones appear to be a suitable technology for resupplying ground delivery vehicles with newly available orders because of their low investment costs, reduced carbon footprint, ability to perform autonomous deliveries, as well as fast point-to-point travel capabilities (Gaba & Winkenbach, 2022).

Below we summarize the main contributions of this paper:

- We formally describe the VRPRD-DR and develop the first MILP formulation for this problem. This formulation models trucks and drones that perform multiple trips (i.e., returns to the depot) during the delivery horizon, synchronizing (both temporally and spatially) these vehicles for resupply operations, which can occur at particular customer locations.
- We develop a unified matheuristic approach that is effective for both the truck-and-drone and the truck-only versions of the VRPRD-DR. For both versions of the problem, the matheuristic outperforms state-of-the-art solvers for instances of 15 or more customers.
- For instances of up to 100 customers, we show the benefits of drone resupply to reduce completion times and the number of times the trucks need to return to the depot. We also provide several managerial insights regarding consolidation and fleet utilization.

The rest of this paper is structured as follows. Related works to the VRPRD-DR are reviewed in Section 2. In Section 3, we formally define the VRPRD-DR and present the MILP formulation. The matheuristic approach is described in Section 4. Section 5 contains the computational results of the MILP model and the matheuristic. Finally, concluding remarks and future research directions are discussed in Section 6.

2. Literature review

The VRPRD-DR can be classified within the family of routing problems with release dates. These problems mainly arise when logistics companies must deliver orders that become ready for dispatch throughout the day, and are found in make-to-order operations (Pina-Pardo et al., 2021), last-mile logistics (Mor & Speranza, 2020), and same-day delivery (Klapp et al., 2018).

To the best of our knowledge, Cattaruzza et al. (2016a) were the first to introduce the concept of release dates in routing problems. They extended the Vehicle Routing Problem (VRP) with time windows to model vehicles that return to the depot multiple times to collect newly released orders. Their computational results showed the efficiency of a proposed memetic algorithm. The first MILP formulation for the problem, which considers a single delivery vehicle, was proposed by Archetti et al. (2018). Similar to Cattaruzza et al. (2016a), the strategy proposed by these authors is for the vehicle to return to the depot periodically during the day to pick up new orders. Archetti et al. (2018) proposed an iterative local search heuristic and evaluated its effectiveness against optimal solutions obtained from a MILP model. Li et al. (2020) presented a MILP formulation and an adaptive neighborhood search heuristic to find the set of delivery routes for a fleet of vehicles to minimize the total completion time of the delivery process. Zhen et al. (2020) developed a MILP model and a hybrid particle swarm algorithm for a problem that considers multiple vehicles, multiple depots, time windows, and order release dates. Archetti et al. (2015) and Reyes et al. (2018) both study the complexity of routing problems with release dates. None of these, however, consider using drones.

A recent application of routing problems with release dates is Same-Day Delivery (SDD) services, where providers must prepare and deliver orders on the same day they are placed. Thus, in this case, the orders and their timing are unknown at the beginning of the day. While SDD has attracted increased attention in the literature over the past few years, the most related works to the VRPRD-DR are Klapp et al. (2018), Ulmer and Thomas (2018), Ulmer et al. (2019), and Chen et al. (2022). Assuming a single delivery vehicle, Klapp et al. (2018) studied the trade-off between waiting for additional products (for greater consolidation) and dispatching the vehicle as soon as possible (for faster delivery). The authors developed a Markov Decision Process (MDP), where decisions are made at predefined moments of the day. They also proposed a MILP formulation for the deterministic case. Ulmer et al. (2019) studied the benefits of preemptive depot returns, which allow vehicles to return to the depot to retrieve new orders before finishing their current delivery routes. They defined the problem as an MDP and presented an approximate value iteration procedure and a routing heuristic approach to reduce the size of both the state and decision spaces. Ulmer and Thomas (2018) proposed combining conventional vehicles and drones for SDD, where drones are employed to make autonomous deliveries from the depot to the customer. They developed an assignment and routing heuristic based on node insertions used with a Policy Function Approximation (PFA) to decide whether an order is to be delivered by a drone or by a conventional vehicle. Finally, extending Ulmer and Thomas (2018), Chen et al. (2022) developed a deep Q-learning solution approach to estimate (i.e., learn) the value of taking an action in a given state. The authors used the assignment and routing heuristics from Ulmer and Thomas (2018) to update the planned routes for trucks and drones. In their computational results, Chen et al. (2022) demonstrated the effectiveness of the deep Q-learning approach over the PFA proposed by Ulmer and Thomas (2018). Note that these works did not consider drones working in tandem with ground vehicles.

In addition to its relationship with routing problems with release dates, the VRPRD-DR belongs to the drone routing literature. As highlighted in Section 1, most papers have proposed using drones to deliver directly to consumers, even though the technology and protocols for such operations are not available in most real-world settings. The contributions have mainly focused on extending the truck-and-drone delivery system proposed by Murray and Chu (2015) to consider multiple trucks and drones (Kitjacharoenchai et al., 2019; Nguyen et al., 2022; Saleu et al., 2022; Schermer et al., 2019b; Yin et al., 2023), drone stations (Chauhan et al., 2019; Kloster et al., 2023; Rave et al., 2023; Schermer et al., 2019), and drones with a payload capacity greater than one parcel (Gonzalez-R et al., 2020; Poikonen & Golden, 2020; Yanpirat et al., 2023). However, most of these have assumed that drones can seamlessly interact with consumers and that all orders are available for dispatch at the beginning of the planning horizon (Macrina et al., 2020), two major assumptions that we do not make in this paper.

Although drone resupply represents a potentially effective strategy to eliminate line-hauls and back-hauls of delivery vehicles from and to the depot to dispatch and collect new orders (Moshref-Javadi & Winkenbach, 2021), to the best of our knowledge, using drones for sending newly available orders to vehicles has been studied only four times in the literature to date. Dayarian et al. (2020) were the first to propose this use of drones, considering a parcel delivery system

composed of a single delivery truck and a single drone. Without proposing an exact solution method, the authors used a heuristic to compare the use of drone resupply with a traditional truck-only delivery system. McCunney and Cauwenberghe (2019) introduced a two-echelon delivery network where drones are used to send products to urban transshipment points which are in turn collected by vehicles for last-mile delivery. Using a simulation approach, these authors concluded that the proposed delivery system reduces truck travel and delivery times. Pina-Pardo et al. (2021) developed the first MILP formulation for a VRP with drone resupply, which considers a single delivery truck that receives newly released orders via a single drone (they did not consider truck returns to the depot in their main analysis). Compared to the traditional approach proposed by Archetti et al. (2018), where the truck periodically returns to the depot to pick up new orders, the authors showed that drone resupply reduces completion times under a wide variety of parameters. Finally, Dienstknecht et al. (2022) studied the benefits of drone resupply for the problem considering a single capacitated truck and a single drone capable of carrying one order per sortie. In this setting, the truck leaves the depot with a subset of orders, and the drone resupplies the rest of the orders one by one. Since the solution approaches proposed by Dienstknecht et al. (2022) assumes that the truck can only perform a single route during the day, their numerical experiments showed that truck returns to the depot are still needed, due mainly to the small drone capacity and resulting truck waiting times for the drone.

Considering the extant literature, there are several pertinent research gaps. In both deterministic and stochastic vehicle routing settings, the approach commonly employed in the literature has been to instruct an en route vehicle to return to the depot to collect any newly released orders (Archetti et al., 2018; Klapp et al., 2018; Ulmer et al., 2019). On the other hand, nearly every drone paper has assumed that all orders are available for dispatch at the beginning of the planning horizon (Macrina et al., 2020) and have also assumed that drones can safely and effectively interact with customers (Boysen et al., 2021; Gaba & Winkenbach, 2020). We dispose of all of these assumptions. This paper uses as a base the work of Pina-Pardo et al. (2021) to consider the quite significant extensions of multiple trucks and multiple drones, where newly released orders can be either collected by the trucks at the depot or resupplied via drones. Specifically, we model trucks and drones that perform multiple trips (i.e., returns to the depot) during the delivery horizon, synchronizing these vehicles spatially and temporally for resupply operations. These features are widely recognized as challenging in the vehicle routing literature due to the uncertain number of trips that vehicles must execute and the coordination required between trucks and drones (see, e.g., Archetti et al., 2018; Cattaruzza et al., 2016b; Dellaert et al., 2019; Drexl, 2012). Furthermore, the drones are not dedicated to any particular truck (i.e., drones can resupply any truck throughout the delivery horizon), a relaxation of the common assumption of most drone delivery papers (Moshref-Javadi & Winkenbach, 2021).

3. The vehicle routing problem with release dates and drone resupply

In this section, we formally describe the VRPRD-DR and present the MILP model. Then, we devise additional valid inequalities to improve the computational performance of the MILP model.

3.1. Problem description

In the VRPRD-DR, we consider multiple delivery trucks and resupply drones working in tandem. We assume that trucks are uncapacitated, but drones have limited flight endurance and load capacity. In addition, we assume that customer locations and their order release times are known at the time of planning. We consider that there is a service time to serve each customer. We also consider that each order requires certain units of drone payload capacity.

To collect newly released orders, the trucks (i) can resupply themselves by returning to the depot (at any time and at any point on their routes) or (ii) can be resupplied by dispatching drones from the depot. In the case of drone resupply, we assume that the drivers take a fixed amount of time to receive the drones. We also consider a fixed amount of time to receive a vehicle (truck or drone) at the depot and prepare it for another trip. For example, in the case of a drone, this time may include the time to land, interchange its battery, and load any available order that is assigned to the drone. Furthermore, we restrict drone resupply to occur at customer locations and for each drone to make only one rendezvous per sortie from the depot. This is a reasonable assumption, and analogous to the ones usually used in the literature for truck-drone tandems (Macrina et al., 2020). Finally, we consider that drones will always arrive at the same time or after the trucks (i) arrive at the meeting location and (ii) serve the corresponding customer (if the trucks have the customer's order already loaded). In other words, a drone waiting for a truck at the rendezvous location is unnecessary since, without loss of optimality, its dispatch from the depot can always be delayed to arrive by the time the drivers are ready to receive the drone. Note that these assumptions, except perhaps the single rendezvous per resupply, apply to any type of resupply vehicle, such as cargo bikes or ground robots. Therefore, the MILP model and the matheuristic presented below could be used for any vehicle type combination as long as these assumptions hold.

In the following, we define a *tour* as a sequence of customers visited by a given truck, which starts and ends at the depot without an intermediate stop at the depot. A drone round trip flight from the depot is called a *sortie*. In the VRPRD-DR, trucks can perform multiple tours and drones fly multiple sorties during the delivery horizon. Further, we define a *master tour* as the complete sequence of customers assigned to a given truck. Note that a feasible VRPRD-DR solution defines exactly one master tour for each truck, and these master tours are mutually exclusive and collectively exhaustive. Finally, the set of all master tours assigned to the trucks is referred to as a *VRP solution*.

The VRPRD-DR contains two types of decisions that must be made simultaneously and must be synchronized with respect to time and space: routing and loading. Fig. 1 shows an illustrative example of a feasible VRPRD-DR solution for 12 customers, considering two trucks and two drones. Fig. 1(a) shows the tours assigned to each truck. Specifically, Truck 1 performs two tours $(1,2,3,4,5)$ and $(6,7,8)$ that make up its master tour $(1,2,3,4,5,6,7,8)$, while Truck 2 performs a single tour that forms its master tour $(9,10,11,12)$. Fig. 1(b) depicts the loading decisions. The white nodes represent the customers whose orders were loaded at the depot before the start of the first tour of each truck, the light gray nodes are the customers whose orders were collected by the trucks when they returned to the depot, and the dark gray nodes are those customers whose orders were resupplied by drone sorties. The black dashed lines represent the sorties of Drone 1, while the black dotted lines represent the sortie of Drone 2. Additionally, the continuous black lines highlight the depot return of Truck 1, performed after serving Customer 5 and before visiting Customer 6 (note that trucks can return to the depot regardless of whether they are empty or have orders loaded). The label of each line shows the subset of orders to be collected by or resupplied to each truck at each loading operation. For example, Drone 1 resupplies Orders 3 and 4 at Node 3 on its first sortie, and Orders 11 and 12 at Node 10 on its second sortie. Additionally, Truck 1 collects the order of Customer 7 when it returns to the depot after serving Customer 5. Fig. 1(b) shows that, for each drone sortie, the loading decisions involve selecting the subset of orders that will be carried on the drone and the rendezvous location with the truck. Additionally, the timing of each rendezvous must also be coordinated between the truck and the drone. Note that the drones are not dedicated to any particular truck. Finally, as Fig. 1(b) shows,

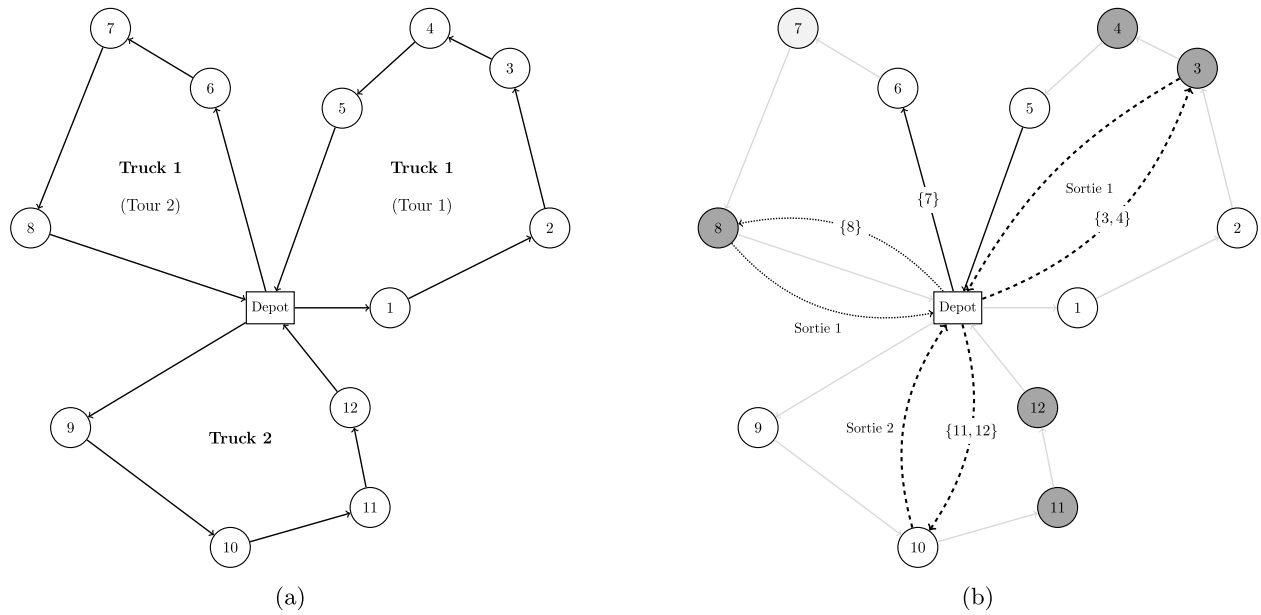

Fig. 1. Illustrative example. (a) Truck tours. (b) Loading operations.

loading decisions require choosing the timing and orders to be collected through depot returns (if any) by each truck.

To be consistent with the research studies on both drone delivery and routing problems with release dates (Archetti et al., 2018; Chung et al., 2020; Mor & Speranza, 2020), we aim to minimize the completion time (i.e., the makespan) of the delivery service, which is the time when the last truck returns to the depot (from its last tour). This metric allows for better distribution of driver workload (Bertazzi et al., 2015) and adequately balances the wait for new orders and the truck travel times. Focusing solely on travel times, for example, would mean waiting for all orders and then performing an optimal Traveling Salesman Problem (TSP) tour, which can lead to unreasonably late deliveries and/or long working hours.

3.2. Mathematical formulation

Let $N = \{1, 2, \dots, n\}$ be the set of customers, V the set of delivery trucks, and K the set of drones. Let $D \subseteq N$ represent the set of nodes that drones can visit according to flight endurance. We denote the single depot with the subindices 0 and $n+1$. Additionally, let $D_0 = D \cup \{0\}$, $D_{n+1} = D \cup \{n+1\}$, and $N_{0,n+1} = N \cup \{0, n+1\}$ be the sets of nodes that include the depot. The problem is defined on the directed graph $G = (N_{0,n+1}, A)$, where $A = \{(i,j) \in N_0 \times N_{n+1} : i \neq j\}$ is the set of all arcs between the nodes.

To streamline the MILP formulation, we define the sets $W = \{(i,j) \in D_0 \times D_{n+1} : i \neq j\}$, $Y = D \times N$ and $\hat{Y} = N \times N$. These sets are also used to simplify the exposition of the MILP formulations presented in Sections 4.2 and 4.5. The parameters of the MILP formulation are described in Table 1, while the decision variables are listed in Table 2.

Let us further explain the meaning of some decision variables before presenting the MILP formulation. We interpret $x^v_{ij} = 1$ and $\vartheta^v_j = 0$ as the decision to instruct truck $v \in V$ to visit customer $j \in N$ *immediately* after visiting customer $i \in N$ (i.e., without returning to the depot in between). Alternately, $x^v_{ij} = 1$ and $\vartheta^v_j = 1$ denote that truck $v \in V$ returns to the depot after visiting customer $i \in N$ and before visiting customer $j \in N$. Note that, unlike the modeling approach proposed by Neira et al. (2020) using arc-based auxiliary decision variables, we model truck return trips to the depot through the node-based decision variables ϑ^v_j. Preliminary experiments showed that the MILP model presented below decisively outperforms the Neira et al. (2020) arc-based modeling approach. Finally, we interpret $u^k_0 = 1$ as the decision to use drone $k \in K$ to resupply any truck at least once.

Table 1
Parameters used in the MILP model.

Symbol	Description
r_i	Release date of the order of customer $i \in N$.
s_i	Service time at customer $i \in N$.
t_{ij}	Minimum truck travel time between nodes i and j, with $(i,j) \in A$.
d_i	Drone flying time between the depot and customer $i \in N$.
δ_{m}	Time needed by a truck to receive a drone en route.
δ_{d}	Time needed to receive a drone at the depot and prepare it for another sortie.
δ_{t}	Time needed to receive a truck at the depot and prepare it for another tour.
Δ_{ij}	Additional time needed to perform a truck depot return between customers $i \in N$ and $j \in N$, i.e., $\Delta_{ij} = t_{i,n+1} + \delta_{\mathrm{t}} + t_{0j} - t_{ij}$.
q_i	Drone load capacity units required to transport the order of customer $i \in N$.
Q	Drone maximum load capacity per sortie.
M	A sufficiently large constant.

3.2.1. MILP model

The MILP model is presented below.

$$\text{minimize} \quad \mathcal{T} \tag{1}$$

subject to:

Truck master tours:

$$\sum_{v \in V} z^v_i = 1 \qquad \forall i \in N \tag{2}$$

$$\sum_{j \in N} x^v_{0j} = 1 \qquad \forall v \in V \tag{3}$$

$$\sum_{i \in N} x^v_{i,n+1} = 1 \qquad \forall v \in V \tag{4}$$

$$\sum_{j:(i,j) \in A} x^v_{ij} = z^v_i \qquad \forall v \in V, i \in N \tag{5}$$

$$\sum_{i:(i,j) \in A} x^v_{ij} = z^v_j \qquad \forall v \in V, j \in N \tag{6}$$

Drone sorties:

$$\sum_{k \in K} u^k_i \leq 1 \qquad \forall i \in D \tag{7}$$

Table 2
Decision variables used in the MILP model.

Symbol	Description
Truck routing variables	
z_i^v	1, if customer $i \in N$ is served by truck $v \in V$. 0, otherwise.
ϑ_i^v	1, if truck $v \in V$ returns to the depot immediately before serving customer $i \in N$. 0, otherwise.
x_{ij}^v	1, if truck $v \in V$ visits node j after node i, with a possible return to the depot in between, with $(i,j) \in A$. 0, otherwise.
Drone routing variables	
u_i^k	1, if node $i \in D_0$ is visited by drone $k \in K$. 0, otherwise.
w_{ij}^k	1, if drone $k \in K$ flies to customer i, returns to the depot, and then starts a new sortie (not necessarily immediately) towards customer j, with $(i,j) \in W$ such that $i \neq 0$ and $j \neq n+1$. 0, otherwise.
w_{0j}^k	1, if drone $k \in K$ performs the first sortie to customer j, with $(0,j) \in W$. 0, otherwise.
$w_{i,n+1}^k$	1, if drone $k \in K$ performs the final sortie to customer i, with $(i,n+1) \in W$. 0, otherwise.
Loading variables	
γ_i^v	1, if the order for customer $i \in N$ is loaded onto truck $v \in V$ at the depot before starting its first tour. 0, otherwise.
$\hat{y}_{ij}^v$	1, if the order for customer $j \in N$ is collected by truck $v \in V$ at the depot before it initiates a new tour visiting customer $i \in N$. 0, otherwise.
y_{ij}^k	1, if the order for customer $j \in N$ is loaded onto a truck at customer $i \in D$ via drone $k \in K$. 0, otherwise.
Time variables	
T_i	Time when the assigned truck departs customer $i \in N$.
τ_0^v	Time when truck $v \in V$ leaves the depot to perform its first tour.
τ_{n+1}^v	Time when truck $v \in V$ arrives at the depot from its last tour.
$\mathcal{T}$	Completion time of the delivery process.

$$u_i^k \leq u_0^k \qquad \forall k \in K, i \in D \tag{8}$$

$$\sum_{j \in D} w_{0j}^k = u_0^k \qquad \forall k \in K \tag{9}$$

$$\sum_{i \in D} w_{i,n+1}^k = u_0^k \qquad \forall k \in K \tag{10}$$

$$\sum_{j:(i,j) \in W} w_{ij}^k = u_i^k \qquad \forall k \in K, i \in D \tag{11}$$

$$\sum_{i:(i,j) \in W} w_{ij}^k = u_j^k \qquad \forall k \in K, j \in D \tag{12}$$

Loading constraints:

$$\sum_{j:(i,j) \in Y} y_{ij}^k \cdot q_j \leq Q \cdot u_i^k \qquad \forall k \in K, i \in D \tag{13}$$

$$\sum_{j:(i,j) \in Y} y_{ij}^k \geq u_i^k \qquad \forall k \in K, i \in D \tag{14}$$

$$\sum_{v \in V} \sum_{i:(i,j) \in \hat{Y}} \hat{y}_{ij}^v + \sum_{k \in K} \sum_{i:(i,j) \in Y} y_{ij}^k + \sum_{v \in V} \gamma_j^v = 1 \qquad \forall j \in N \tag{15}$$

$$\gamma_j^v \leq z_j^v \qquad \forall j \in N, v \in V \tag{16}$$

$$\sum_{k \in K} y_{ij}^k + z_i^v - 1 \leq z_j^v \qquad \forall (i,j) \in Y, v \in V \tag{17}$$

$$z_j^v + z_i^v \geq 2 \cdot \hat{y}_{ij}^v \qquad \forall (i,j) \in \hat{Y}, v \in V \tag{18}$$

$$\hat{y}_{ij}^v \leq \vartheta_i^v \qquad \forall (i,j) \in \hat{Y}, v \in V \tag{19}$$

$$\sum_{j:(i,j) \in \hat{Y}} \hat{y}_{ij}^v \geq \vartheta_i^v \qquad \forall v \in V, i \in N \tag{20}$$

$$1 - x_{0j}^v \geq \vartheta_j^v \qquad \forall v \in V, j \in N \tag{21}$$

Synchronization and timing constraints:

$$T_j \geq T_i + t_{ij} + s_j - M\left(1 - \sum_{k \in K} y_{ij}^k\right) \qquad \forall (i,j) \in Y \tag{22}$$

$$T_j \geq T_i + t_{ij} + s_j - M\left(1 - \sum_{v \in V} \hat{y}_{ij}^v\right) \qquad \forall (i,j) \in \hat{Y} \tag{23}$$

$$T_j \geq T_i + t_{ij} + \Delta_{ij}\left(\sum_{v \in V} \vartheta_j^v\right) + s_j - M\left(1 - \sum_{v \in V} x_{ij}^v\right) \qquad \forall i \in N, j \in N \backslash (D \cup \{i\}) \tag{24}$$

$$T_j \geq T_i + t_{ij} + \Delta_{ij}\left(\sum_{v \in V} \vartheta_j^v\right) + s_j + \delta_{\mathrm{m}}\left(\sum_{k \in K} u_j^k\right) - M\left(1 - \sum_{v \in V} x_{ij}^v\right) \qquad \forall i \in N, j \in D \backslash \{i\} \tag{25}$$

$$T_j \geq \tau_0^v + t_{0j} + s_j - M(1 - x_{0j}^v) \qquad \forall j \in N \backslash D, v \in V \tag{26}$$

$$T_j \geq \tau_0^v + t_{0j} + s_j + \delta_{\mathrm{m}}\left(\sum_{k \in K} u_j^k\right) - M(1 - x_{0j}^v) \qquad \forall j \in D, v \in V \tag{27}$$

$$\tau_{n+1}^v \geq T_i + t_{i,n+1} - M(1 - x_{i,n+1}^v) \qquad \forall i \in N, v \in V \tag{28}$$

$$T_j \geq T_i - s_i\left(\sum_{k \in K} y_{ii}^k\right) + d_i + \delta_{\mathrm{d}} + d_j + \delta_{\mathrm{m}} + s_j\left(\sum_{k \in K} y_{jj}^k\right) - M\left(1 - \sum_{k \in K} w_{ij}^k\right) \qquad \forall i \in D, j \in D \backslash \{i\} \tag{29}$$

Timing constraints based on order release times:

$$T_i \geq (r_j + d_i + \delta_{\mathrm{m}})\left(\sum_{k \in K} y_{ij}^k\right) + s_i\left(\sum_{k \in K} y_{ii}^k\right) \qquad \forall (i,j) \in Y \tag{30}$$

$$\tau_0^v \geq r_i \cdot \gamma_i^v \qquad \forall v \in V, i \in N \tag{31}$$

$$T_i \geq (r_j + t_{0i} + s_i)\left(\sum_{v \in V} \hat{y}_{ij}^v\right) \qquad \forall i \in N \backslash D, j \in N \tag{32}$$

$$T_i \geq (r_j + t_{0i} + s_i)\left(\sum_{v \in V} \hat{y}_{ij}^v\right) + \delta_{\mathrm{m}}\left(\sum_{k \in K} u_i^k\right) \qquad \forall i \in D, j \in N \tag{33}$$

Definition of the completion time:

$$\mathcal{T} \geq \tau_{n+1}^v \qquad \forall v \in V \tag{34}$$

Non-negativity and integrality constraints:

$$x_{ij}^v \in \{0,1\} \qquad \forall (i,j) \in A, v \in V \tag{35}$$

$$z_i^v, \gamma_i^v, \vartheta_i^v \in \{0,1\} \qquad \forall i \in N, v \in V \tag{36}$$

$$w_{ij}^k \in \{0,1\} \qquad \forall (i,j) \in W, k \in K \tag{37}$$

$$u_i^k \in \{0,1\} \qquad \forall i \in D_0, k \in K \tag{38}$$

$$y_{ij}^k \in \{0,1\} \qquad \forall (i,j) \in Y, k \in K \tag{39}$$

$$\hat{y}_{ij}^v \in \{0,1\} \qquad \forall (i,j) \in \hat{Y}, v \in V \tag{40}$$

$$T_i \geq 0 \qquad \forall i \in N \tag{41}$$

$$\tau_0^v, \tau_{n+1}^v \geq 0 \qquad \forall v \in V \tag{42}$$

$$\mathcal{T} \geq 0 \tag{43}$$

The objective function (1) minimizes the total completion time of the delivery process. Constraints (2) force each customer to be visited by exactly one truck. Constraints (3)–(6) define the truck flow balance

for each node. Constraints (7) state that each customer can be visited by at most one drone. Constraints (8) indicate that a drone can visit a node as long as it is used to resupply a truck at least once. Constraints (9)–(12) define the drone flow balance for each node. The payload size of each drone at each resupply operation is restricted in (13). Constraints (14) state that drone $k \in K$ must resupply at least one order if it visits customer $i \in N$. Constraints (15) force each order to be loaded onto exactly one truck through exactly one loading operation: depot return, drone resupply, or at the depot before a truck begins its first tour. Constraints (16)–(18) ensure that each order must be delivered by the truck on which it was loaded. In particular, Constraints (17) force truck $v \in V$ to serve customer $j \in N$ if the order of this customer is transferred to this truck via a drone at the location of customer $i \in N$ (i.e., if $z_i^v = 1$ and $y_{ij}^k = 1$, for some $k \in K$), while Constraints (18) force the same truck to visit customer $j \in N$ and customer $i \in N$ if the order of customer $j \in N$ is collected by this truck at the depot before it starts a new tour visiting customer $i \in N$. Constraints (19) force a truck to return to the depot if at least one order is collected before visiting customer $i \in N$. Similar to Constraints (14), Constraints (20) state that truck $v \in V$ must collect at least one order if it returns to the depot before visiting customer $i \in N$. Constraint (21) forces depot returns to occur only after the first customer has been visited. Constraints (22) establish that the departure time at customer $j \in N$ must be greater than the departure time at customer $i \in N$ if the order of customer $j \in N$ is loaded onto a truck at customer $i \in N$ via a drone. Similarly, Constraints (23) force the departure time at customer $j \in N$ to be greater than the departure time at customer $i \in N$ if the order of customer $j \in N$ is collected by a truck at the depot before visiting customer $i \in N$. As a whole, Constraints (22)–(33) allow computing the departure time at each node, as well as synchronizing the drone sorties with the delivery tours of each truck. Constraints (24)–(28) manage subtour elimination for the trucks. Specifically, Constraints (25) and (27) include δ_{m} (i.e., the time needed to receive a drone en route) to consider this time whenever a drone resupply operation occurs at customer $j \in N$. Further, Constraints (24) and (25) include Δ_{ij} to consider the extra time needed for a truck to return to the depot between customers $i \in N$ and $j \in N$. Constraints (29) and (30) establish that a drone can initialize a new sortie as long as it has returned to the depot, and all orders that will be resupplied have been released. The departure time of the first tour of each truck from the depot is determined by (31). Constraints (32) and (33) ensure that the departure time from customer $j \in N$ must be greater than the release date of any order collected by a truck at the depot (before visiting customer $j \in N$) plus the truck travel time from the depot to that customer. Constraints (34) define the completion time of the delivery process. Finally, non-negativity and integrality constraints are included in (35)–(43).

3.2.2. Illustrative examples of the synchronization and timing constraints

Fig. 2 depicts an example of the synchronization and timing Constraints (22)–(33) when there is a drone resupply operation planned at customer $j \in N$. After resupplying a truck at customer $\ell \in N$, drone $k \in K$ will transfer orders $o_1, o_2 \in N \backslash \{j\}$ to truck $v \in V$ at customer j (the trucks resupplied by these sorties may be different). The drone may start this new sortie as long as it has returned and has been prepared at the depot (at time $T_\ell + d_\ell + \delta_{\mathrm{d}}$) and orders o_1 and o_2 have been released (at time $\max\{r_{o_1}, r_{o_2}\}$). Truck v may arrive and serve customer j at time $T_i + t_{ij} + s_j$. Therefore, T_j must satisfy the following inequalities, which consider that $x_{ij}^v = 1$, $\vartheta_j^v = 0$ (i.e., there is no depot return immediately before visiting customer j), $u_j^k = 1$, $w_{\ell j}^k = 1$, $y_{jo_1}^k = 1$, $y_{jo_2}^k = 1$, $y_{jj}^k = 0$, and $y_{\ell\ell}^k = 0$:

Inequality:	Constraints:
$T_j \geq T_i + t_{ij} + s_j + \delta_{\mathrm{m}}$	(25)
$T_j \geq T_\ell + d_\ell + \delta_{\mathrm{d}} + d_j + \delta_{\mathrm{m}}$	(29)
$T_j \geq r_{o_1} + d_j + \delta_{\mathrm{m}}$	(30)
$T_j \geq r_{o_2} + d_j + \delta_{\mathrm{m}}$	(30)

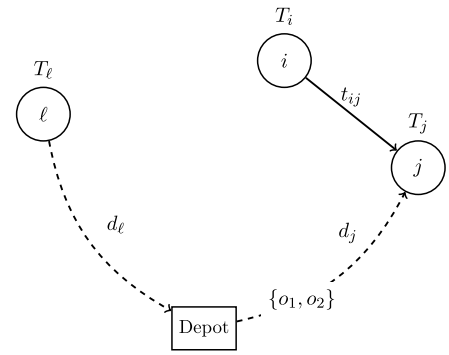

Fig. 2. First example of the synchronization and timing constraints.

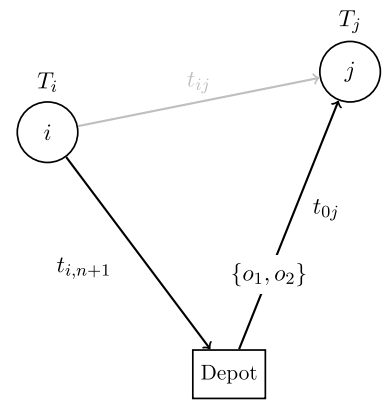

Fig. 3. Second example of the synchronization and timing constraints.

Recall that the MILP formulation (1)–(43) models that drones will always arrive after the truck arrives at the rendezvous location and the corresponding customer is served by the truck (see Section 3.1). Thus, the time to receive a drone en route, δ_{m}, is included in all inequalities since it must be taken into account after customer j is served by the truck and the drone reaches the location of this customer.

In the special case that the drone carries the order of customer j (i.e., $y_{jj}^k = 1$), then the service time of this customer must be included in the last three inequalities, meaning that the customer can only be served after receiving the drone. Further, after receiving the drone and unloading its orders, we consider that the driver sends the drone back to the depot and then serves customer j. Note that this sequence of actions is appropriately addressed in the MILP formulation in Constraints (29) and (30).

A second example is shown in Fig. 3, which considers the case of a truck depot return. After visiting customer $i \in N$, truck $v \in V$ performs a depot return to collect orders $o_1 \in N$ and $o_2 \in N$ and then starts a new tour by visiting customer $j \in N$. The truck may start this new tour as long as it has returned and has been prepared at the depot (at time $T_i + t_{i,n+1} + \delta_{\mathrm{t}}$) and orders o_1 and o_2 have been released (at time $\max\{r_{o_1}, r_{o_2}\}$). Therefore, T_j must satisfy the following inequalities, which consider that $x_{ij}^v = 1$, $\vartheta_j^v = 1$, $u_j^k = 0$, $\hat{y}_{jo_1}^v = 1$, and $\hat{y}_{jo_2}^v = 1$:

Inequality:	Constraints:
$T_j \geq T_i + t_{ij} + \Delta_{ij} + s_j$	(24) and (25)
$T_j \geq r_{o_1} + t_{0j} + s_j$	(32) and (33)
$T_j \geq r_{o_2} + t_{0j} + s_j$	(32) and (33)

3.3. Valid inequalities

To be considered feasible, a VRPRD-DR solution requires that the truck tours be (both temporally and spatially) synchronized with the

possible drone resupply sorties. To this end, we use, in Constraints (22)–(29), a variation of the Miller et al. (1960) subtour elimination constraints that are typically employed in VRPs with time windows (Toth & Vigo, 2014). However, it is well known that using these constraints in VRPs leads to weak linear programming relaxations (Bektaş & Gouveia, 2014), which we corroborated in our initial experiments. An alternative is to lift subtour elimination constraints, as was done by Desrochers and Laporte (1991) for the VRP with time windows. Their lifting procedure aims to find the largest possible value of α_{ji} (i.e., the coefficient of the reverse arc variables) such that subtour elimination constraints remain valid. To do so, Desrochers and Laporte (1991) use the upper bounds of the customer time windows (see Proposition 9 of Desrochers and Laporte (1991)), which is not part of our problem definition (see Section 3.1). Without these bounds, finding valid values of α_{ji} is not straightforward. This is because, in the VRPRD-DR, the trucks might wait (for as long as necessary) for a drone to arrive at a customer location and might return to the depot between two consecutive nodes. Lifting subtour elimination constraints might be an interesting future research effort.

Therefore, extending Pina-Pardo et al. (2021), we derive the following valid inequalities to improve the computational performance of the MILP formulation (1)–(43). Numerical experiments showing the effectiveness of these constraints are presented in Online Appendix A.

3.3.1. Completion time constraints

Let $\epsilon_j^v \in \mathbb{R}_+$ be a decision variable that denotes the elapsed time between the arrival time at customer $j \in N$ if the corresponding truck $v \in V$ travels directly from its predecessor, and the actual departure time of the truck from that customer (which considers the customer service time and any additional time for returning to the depot or receiving a drone at the location of that customer). These variables can be computed through the following inequalities:

$$\epsilon_j^v \geq T_j - (T_i + t_{ij}) - M(1 - x_{ij}^v) \qquad \forall j \in N, i \in N \backslash \{j\}, v \in V \tag{44}$$

$$\epsilon_j^v \geq T_j - (\tau_0^v + t_{0j}) - M(1 - x_{0j}^v) \qquad \forall j \in N, v \in V \tag{45}$$

$$\epsilon_j^v \geq 0 \qquad \forall j \in N, v \in V \tag{46}$$

Thus, the time when truck $v \in V$ returns to the depot can be bounded as follows:

$$\tau_{n+1}^v \geq \tau_0^v + \sum_{(i,j)\in A} t_{ij} x_{ij}^v + \sum_{j \in N} \epsilon_j^v \qquad \forall v \in V \tag{47}$$

3.3.2. Timing constraints

We also include the following valid inequalities:

$$\mathcal{T} \geq T_i + t_{i,n+1} \qquad \forall i \in N \tag{48}$$

$$T_i \geq r_i + \min\{d_i + \delta_{\mathrm{m}}, t_{0i}\} + s_i \qquad \forall i \in N \tag{49}$$

Constraints (48) indicate that the completion time of the delivery process is bounded by the departure time plus the truck travel time from each customer to the depot. Additionally, the departure time at each customer is bounded in Constraints (49). The rationale behind (49) is that the minimum time at which an order can be delivered is given by its release date plus the minimum between the drone flying time (including δ_{m}, the time to receive a drone en route) and the truck travel time, plus the corresponding service time.

Constraints (48) and (49) allow us to introduce the following proposition, which establishes a lower bound for the objective function based on the problem parameters. Later, in Section 5.2, we use Proposition 1 to define a termination criterion for the matheuristic.

Proposition 1. *The inequality*

$$\mathcal{T} \geq \max_{i \in N}\left\{ r_i + \min\{d_i + \delta_m, t_{0i}\} + s_i + t_{i,n+1} \right\} \tag{50}$$

sets a lower bound on the completion time of the delivery process.

Proof. The proof is straightforward. Constraints (48) can be written as follows:

$$\mathcal{T} \geq \max_{i \in N}\left\{ T_i + t_{i,n+1} \right\}$$

Furthermore, since T_i ($\forall i \in N$) is bounded in Constraints (49), we can establish that:

$$\mathcal{T} \geq \max_{i \in N}\left\{ T_i + t_{i,n+1} \right\} \geq \max_{i \in N}\left\{ r_i + \min\{d_i + \delta_{\mathrm{m}}, t_{0i}\} + s_i + t_{i,n+1} \right\}$$

which completes the proof. □

4. Solution approach

As we will show using numerical experiments in Section 5, the MILP model rapidly becomes intractable as the number of customers grows. Therefore, in this section, we propose a matheuristic approach based on a tabu search scheme (see Archetti & Speranza, 2014 for a survey on matheuristics for routing problems). In this approach, we repeatedly modify the VRP solutions and define optimal loading operations (for a given VRP solution) through the Loading Subproblem (LSP), a fast subordinate MILP formulation.

An appealing property of the matheuristic is that it can be used to solve both the truck-and-drone and the truck-only versions of the problem. The difference relies on the loading operations defined by the LSP for a given VRP solution. Specifically, for the truck-and-drone version, the LSP defines both the drone resupplies and the truck depot returns that minimize the completion time of a given VRP solution. On the other hand, the LSP defines the optimal truck returns to the depot for the truck-only version.

In the remainder of this section, we provide a detailed description of the matheuristic. We begin by presenting an overview of the matheuristic through a flowchart. We then describe the LSP and the two pruning strategies that are used to identify promising VRP solutions. The procedures for generating the initial solution and neighborhoods are then described, as well as the mechanisms for updating the next solution and the tabu list. Finally, we discuss the (minor) modifications needed for using the matheuristic to solve the truck-only version of the problem.

4.1. Matheuristic approach

A flowchart of the matheuristic is presented in Fig. 4. We start with an initial VRP solution obtained through an adapted version of the well-known I1 insertion heuristic (Solomon, 1987). In the main iterative process, we create neighboring solutions through a *Relocate* operator and compute a lower bound to initially evaluate solution quality. For each neighbor whose lower bound satisfies a threshold, we invoke the LSP with cutoff information, corresponding to the best-so-far solution's objective value. This allows the early termination of the MILP solver in case it corroborates that no solution of the branch-and-bound tree will improve the best-so-far solution. If the LSP terminates (i.e., runs to optimality), we update the best-so-far solution with the optimal LSP solution. After exploring the entire neighborhood of the current VRP solution, we set the VRP solution for the next iteration and update the tabu list. Furthermore, we apply a diversification mechanism when a criterion is met. The matheuristic terminates by returning the best-so-far solution.

Let us introduce the following additional notation before explaining each of the matheuristic's components. Let $\mathcal{X} = (\mathcal{X}_v)_{v \in V}$ be a VRP solution, where $\mathcal{X}_v = (\mathcal{X}_1^v, \ldots, \mathcal{X}_{m_v}^v)$ is the master tour assigned to truck $v \in V$. We denote by $z(\mathcal{X})$ the optimal value obtained after solving the LSP($\mathcal{X}$) (i.e., the LSP with $\mathcal{X}$ fixed). Finally, let $\mathcal{X}^*$ represent the best VRP solution found so far by the matheuristic.

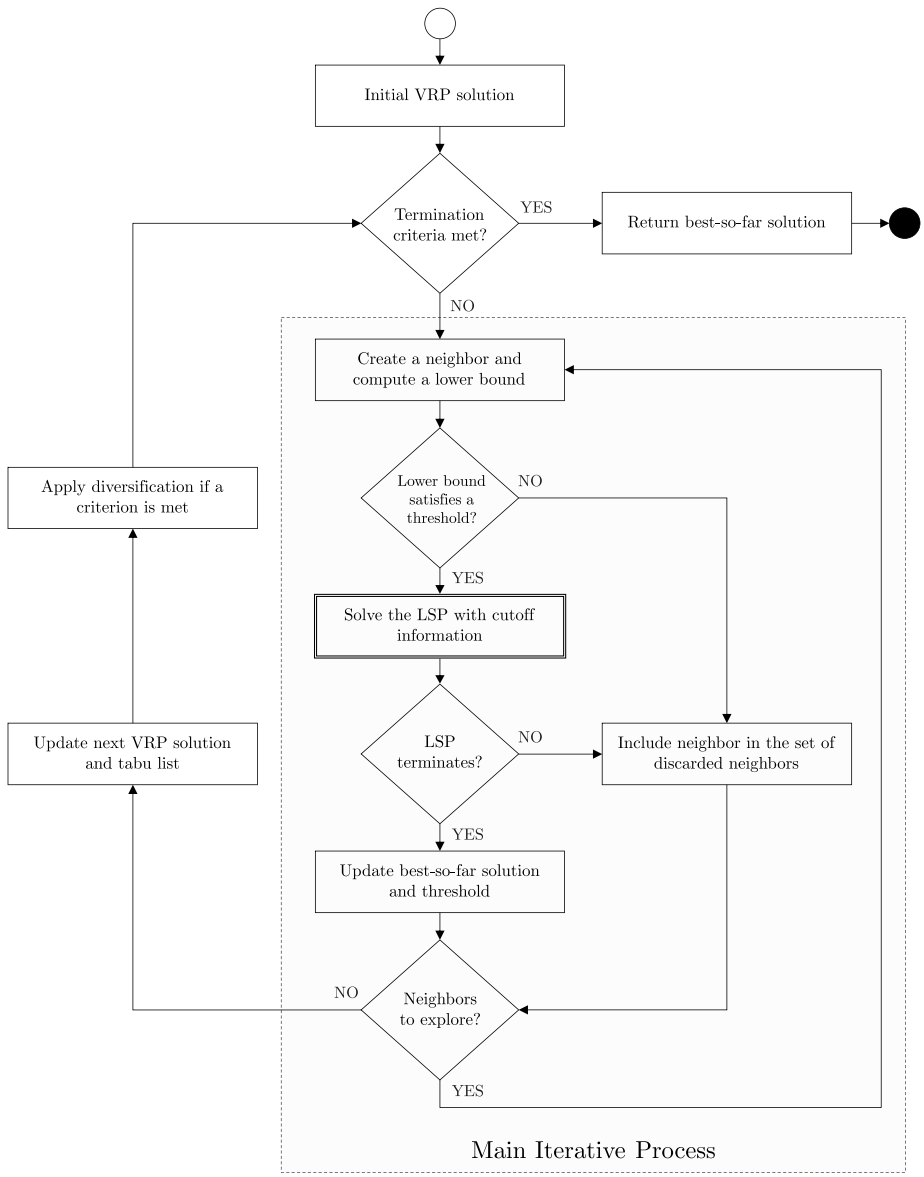

Fig. 4. Matheuristic approach.

4.2. The loading subproblem

The core component of the matheuristic is the LSP, which determines the optimal loading decisions (i.e., drone resupplies and truck depot returns) for a given VRP solution. To do so, in principle, the same MILP formulation (1)–(43) could be used with variables $\boldsymbol{x} = (x_{ij}^v)$ and $\boldsymbol{z} = (z_i^v)$ fixed. However, in our preliminary experiments, we found that the resulting MILP model quickly becomes intractable, mainly because it contains many redundant variables and constraints. Therefore, we develop a more compact MILP formulation based on the observations below. To avoid confusion, we denote $\tilde{x}_{ij}^v$ and $\tilde{z}_i^v$ as the fixed values of the decision variables x_{ij}^v and z_i^v, respectively, which are given by the VRP solution $\mathcal{X}$.

1. *Each order must be loaded before or at the time that the corresponding customer is visited*:
   Let $i \prec j$ denote that customer $i \in N$ precedes customer $j \in N$ in a master tour in the VRP solution $\mathcal{X}$. Similarly, $i \preceq j$ denotes that i precedes j or i is the same as j. According to $\mathcal{X}$, we introduce the following sets:
   $$Y(\mathcal{X}) = \left\{(i,j) \in Y : \exists v \in V, (\tilde{z}_i^v = 1 \wedge \tilde{z}_j^v = 1) \wedge i \preceq j\right\},$$
   $$\hat{Y}(\mathcal{X}) = \left\{(i,j) \in \hat{Y} : \exists v \in V, (\tilde{z}_i^v = 1 \wedge \tilde{z}_j^v = 1) \wedge i \preceq j\right\},$$
   as subsets of Y and $\hat{Y}$ (see Section 3.2), respectively. The definition of $Y(\mathcal{X})$ allows us to define the variables $\boldsymbol{y} = (y_{ij}^k)$ only for $(i,j) \in Y(\mathcal{X})$ and $k \in K$, as well as to remove Constraints (22). Similarly, we define $\hat{\boldsymbol{y}} = (\hat{y}_{ij}^v)$ only for $(i,j) \in \hat{Y}(\mathcal{X})$ and $v \in V$ and remove Constraints (23).
2. *Orders must be loaded onto the truck that delivers them*:
   This observation allows us to remove the vehicle index from the variables $\boldsymbol{\gamma} = (\gamma_i^v)$, $\boldsymbol{\vartheta} = (\vartheta_i^v)$, and $\hat{\boldsymbol{y}}$, as well as to eliminate Constraints (16)–(18) and (21).
3. *The sequence of drone resupply operations must be consistent with routing*:
   As an example, consider the tour $(1,2,3,4,5)$ illustrated in Fig. 1(b). If drone $k \in K$ visits customer 3, it cannot fly to customers 1 and 2 later. To restrict these operations, we introduce:
   $$W(\mathcal{X}) = \left\{(i,j) \in W : \exists v \in V, (\tilde{z}_i^v = 1 \wedge \tilde{z}_j^v = 1) \Rightarrow i \prec j\right\}$$
   to define $\boldsymbol{w} = (w_{ij}^k)$ only for $(i,j) \in W(\mathcal{X})$ and $k \in K$.

4.2.1. MILP formulation

Replacing the sets Y, $\hat{Y}$ and W by $Y(\mathcal{X})$, $\hat{Y}(\mathcal{X})$ and $W(\mathcal{X})$ in the MILP model described in Section 3, respectively, the MILP model for the LSP($\mathcal{X}$) is given below.

$$z(\mathcal{X}) = \min_{\substack{\boldsymbol{y},\hat{\boldsymbol{y}},\boldsymbol{\gamma},\boldsymbol{\vartheta},\\ \boldsymbol{w},\boldsymbol{u},\boldsymbol{\tau},\boldsymbol{T},\mathcal{T}}} \{\mathcal{T} : (7)\text{–}(14), (29), (30), (34), (37)\text{–}(39), (41)\text{–}(43), (48), (52)\text{–}(65)\} \tag{51}$$

The set of Constraints (52)–(65) is defined as follows.

Modification of Constraints (15), (19), and (20):

$$\sum_{i:(i,j)\in\hat{Y}(\mathcal{X})} \hat{y}_{ij} + \sum_{k\in K}\sum_{i:(i,j)\in Y(\mathcal{X})} y^k_{ij} + \gamma_j = 1 \qquad \forall j \in N \tag{52}$$

$$\hat{y}_{ij} \le \vartheta_i \qquad \forall (i,j) \in \hat{Y}(\mathcal{X}) \tag{53}$$

$$\sum_{j:(i,j)\in\hat{Y}(\mathcal{X})} \hat{y}_{ij} \ge \vartheta_i \qquad \forall i \in N \tag{54}$$

Modification of timing Constraints (24)–(28):

$$T_j \ge T_i + t_{ij} + s_j + \Delta_{ij}\cdot\vartheta_j \qquad \forall i \in N, j \in N\backslash(D\cup\{i\}) : \exists v \in V, \tilde{x}^v_{ij} = 1 \tag{55}$$

$$T_j \ge T_i + t_{ij} + s_j + \Delta_{ij}\cdot\vartheta_j + \delta_{\mathrm{m}}\left(\sum_{k\in K} u^k_j\right) \qquad \forall i \in N, j \in D\backslash\{i\} : \exists v \in V, \tilde{x}^v_{ij} = 1 \tag{56}$$

$$T_j \ge \tau^v_0 + t_{0j} + s_j \qquad \forall v \in V, j \in N\backslash D : \tilde{x}^v_{0j} = 1 \tag{57}$$

$$T_j \ge \tau^v_0 + t_{0j} + s_j + \delta_{\mathrm{m}}\left(\sum_{k\in K} u^k_j\right) \qquad \forall v \in V, j \in D : \tilde{x}^v_{0j} = 1 \tag{58}$$

$$\tau^v_{n+1} \ge T_i + t_{i,n+1} \qquad \forall v \in V, i \in N : \tilde{x}^v_{i,n+1} = 1 \tag{59}$$

Modification of timing Constraints (31)–(33):

$$\tau^v_0 \ge r_i\cdot\gamma_i \qquad \forall v \in V, i \in N : \tilde{z}^v_i = 1 \tag{60}$$

$$T_i \ge (r_j + t_{0i} + s_i)\cdot\hat{y}_{ij} \qquad \forall (i,j) \in \hat{Y}(\mathcal{X}) : i \in N\backslash D \tag{61}$$

$$T_i \ge (r_j + t_{0i} + s_i)\cdot\hat{y}_{ij} + \delta_{\mathrm{m}}\left(\sum_{k\in K} u^k_i\right) \qquad \forall (i,j) \in \hat{Y}(\mathcal{X}) : i \in D \tag{62}$$

New integrality constraints:

$$\gamma_i \in \{0,1\} \qquad \forall i \in N \tag{63}$$

$$\hat{y}_{ij} \in \{0,1\} \qquad \forall (i,j) \in \hat{Y}(\mathcal{X}) \tag{64}$$

$$\vartheta_j \in \{0,1\} \qquad \forall j \in N : \forall v \in V, x^v_{0j} = 0 \tag{65}$$

Note that the only big-M constraints in the MILP model (51) are Constraints (29) and the only decisions variables with a truck index are τ^v_0 and τ^v_{n+1}. The resulting reduction of the LSP feasible solution space facilitates the MILP solver to quickly find optimal solutions, which is crucial when using the matheuristic to solve large instances. Finally, note that Constraints (44)–(47) are not included in the MILP model (51) of the LSP since preliminary experiments showed that they do not necessarily improve its solution times.

4.3. Pruning strategies

This section describes two pruning strategies used to solve the LSP to optimality only for those VRP solutions that have a reasonable potential to improve the best-so-far solution.

4.3.1. Lower bound strategy

Let $\mathcal{X} = (\mathcal{X}_v)_{v\in V}$ be a VRP solution, where $\mathcal{X}_v = (\mathcal{X}^v_1, \dots, \mathcal{X}^v_{m_v})$ is the master tour of truck $v \in V$. Let $a_i = r_i + \min\{d_i + \delta_{\mathrm{m}}, t_{0i}\} + s_i$ denote the lower bound on the departure time from customer $i \in N$ derived in Section 3.3.2. Inspired by Nagata et al. (2010), a lower bound $\ell(\mathcal{X}_v)$ for the completion time of the master tour $\mathcal{X}_v$ (denoted as τ^v_{n+1}) can be computed recursively as follows:

$$\ell_1(\mathcal{X}_v) = \max\left\{a_{\mathcal{X}^v_1},\ t_{0,\mathcal{X}^v_1}\right\}$$

$$\ell_i(\mathcal{X}_v) = \max\left\{a_{\mathcal{X}^v_i},\ \ell_{i-1}(\mathcal{X}_v) + t_{\mathcal{X}^v_{i-1},\mathcal{X}^v_i}\right\} \qquad i = 2, \dots, m_v$$

$$\ell(\mathcal{X}_v) = \ell_{m_v}(\mathcal{X}_v) + t_{\mathcal{X}^v_{m_v},n+1}$$

The above definition allows us to introduce the following proposition:

Proposition 2. *Let $\mathcal{X}$ be a VRP solution.* LB($\mathcal{X}$) *defined as*

$$\mathrm{LB}(\mathcal{X}) = \max_{v\in V}\left\{\ell(\mathcal{X}_v)\right\} \tag{66}$$

sets a lower bound on the optimal value of the LSP($\mathcal{X}$).

Proof. Let $\mathcal{T}^*$ be the optimal LSP($\mathcal{X}$) value. From constraint (34), we have that $\mathcal{T}^* \ge \tau^v_{n+1}$, for all $v \in V$. We have shown above that $\tau^v_{n+1} \ge \ell(\mathcal{X}_v)$, for all $v \in V$. The result follows. □

Proposition 2 allows us to discard VRP solutions *a priori* (i.e., without invoking the LSP). To do so, we compare the corresponding lower bound with a Threshold. A VRP solution whose lower bound fails to meet the threshold is discarded, and is included in the set of discarded neighbors that is used to set the VRP solution for the next iteration, as we will explain in Section 4.4.3.

It is worth noting that, in principle, any threshold could be used to prune VRP solutions. The more restrictive the threshold, the fewer LSPs are invoked, and thus the matheuristic can inspect neighborhoods faster. After some preliminary experimentation, we set the Threshold as a convex combination of the optimal LSP value $z(\mathcal{X}^*)$ and the lower bound LB($\mathcal{X}^*$) of the best-so-far solution $\mathcal{X}^*$, that is,

$$\textsc{Threshold} = (1-\varphi)z(\mathcal{X}^*) + \varphi\mathrm{LB}(\mathcal{X}^*),$$

where $\varphi \in [0,1]$. After preliminary experiments, we set $\varphi = \eta_{\mathrm{current}}/\eta_{\max}$, where η_{current} denotes the current number of iterations without improvement and $\eta_{\max}$ the maximum number of iterations without improvement. Thus, the value of φ is increased adaptively as the number of iterations without improvement increases, allowing for a more intense search in regions of the search space that are likely to be promising, with a less intense search in less promising regions.

The definition of the lower bound $\ell(\mathcal{X}_v)$ also allows us to further tighten the valid inequalities (49) as follows:

$$T_{\mathcal{X}^v_i} \ge \ell_i(\mathcal{X}_v) \qquad \forall v \in V, i \in \{1, \dots, m_v\} \tag{67}$$

Inequalities (67) were included in our computational implementation of the LSP($\mathcal{X}$) since, as confirmed through preliminary experiments, they speed up its solution times.

4.3.2. Cutoff strategy

Whenever a VRP solution successfully satisfies the Threshold, we invoke the LSP with additional information about the completion time of the best-so-far solution. Inspired by Schermer et al. (2019b), we set an upper cutoff in the MILP solver – used for solving the LSP – to discard any node of the branch-and-bound tree whose value is greater than the cutoff value. If the MILP solver detects that no node of the branch-and-bound tree meets the cutoff value, it typically (and quickly) declares the model infeasible and does not provide any solution information (see, for example, IBM, 2020 and Gurobi Optimization, 2021).

In the matheuristic approach, we call the MILP solver with the following upper cutoff: $\textsc{CutOff} = z(\mathcal{X}^*) - \varepsilon$. Similar to Huerta-Muñoz et al. (2022), we consider a small non-negative value ε to discard any neighbor with the same objective value as $z(\mathcal{X}^*)$, the optimal value of

the best-so-far solution. Any discarded neighbor is included in the set of discarded neighbors of the current VRP solution (see Section 4.4.3). The inclusion of ε also guarantees that the best-so-far solution will be updated whenever the LSP is solved to optimality.

4.4. Exploration of the VRP solution space

This section describes how the matheuristic explores the VRP solution space. We begin by describing the procedures for generating the initial VRP solution and neighborhoods. We then present the mechanisms for updating the next solution and managing the tabu list.

4.4.1. Initial solution generation

In Section 3.3.2, we showed that there is a lower bound for the departure time at each customer, which can be interpreted as the opening of the customer's time window. This observation motivated us to generate the initial VRP solution using an adapted version of the insertion heuristic I1 of Solomon (1987) (see also Dullaert & Bräysy, 2003 for a concise description of this heuristic). This is commonly used in metaheuristics for routing problems with time windows (Bräysy & Gendreau, 2005). To employ this heuristic, we assume that each truck has a load capacity to serve at most $\lceil |N|/|V| \rceil$ customers (i.e., we assign roughly the same number of customers to each truck). Additionally, we do not consider an upper bound for each customer's time window.

After generating the initial VRP solution and designating it as the best-so-far solution ($\mathcal{X}^*$), the LSP is solved to obtain an initial feasible solution for the VRPRD-DR. The optimal LSP value is saved in $z(\mathcal{X}^*)$, which is used to evaluate the quality of each generated neighbor, as we previously explained in Section 4.3.

4.4.2. Neighborhood generation

The neighborhood of the current VRP solution $\mathcal{X}$ is generated through a variation of the classic *Relocate* operator (Savelsbergh, 1992). Specifically, we create a neighbor by relocating one customer served by the LastTruck (i.e., the truck that completes its master tour last) in exactly one position of exactly one master tour of $\mathcal{X}$. The latter can be the master tour of the same LastTruck (for intra-route moves) or any other master tour (for inter-route moves). The insertion of the customer in the selected master tour is performed between two consecutive nodes so as to minimize the value of the lower bound described in Section 4.3.1. If more than one insertion position attains the minimum value of the lower bound, we select one uniformly at random. Thus, since we create one neighbor based on each customer served by the LastTruck, the neighborhood of the current VRP solution has a cardinality of $\mathcal{O}(|N|)$.

4.4.3. Solution update for next iteration and tabu list

After exploring the entire neighborhood of the current VRP solution, we set the VRP solution for the next iteration ($\mathcal{X}$) as the best-so-far solution ($\mathcal{X}^*$) if the latter was updated in the current iteration. We also set the LastTruck as the truck that returns to the depot last according to the LSP($\mathcal{X}^*$).

It is worth noting that, due to the use of the two pruning strategies, all neighbors of the current VRP solution might be discarded (i.e., no neighbor updates the best-so-far solution). In this case, we set the VRP solution for the next iteration ($\mathcal{X}$) as the best discarded non-tabu neighbor based on the lower bounds of all neighbors. Furthermore, we set the LastTruck as the truck with the largest lower bound $\ell(\mathcal{X}_v)$ (see Eq. (66)). Although the latter is a heuristic choice (since it would be necessary to solve the LSP to identify the last truck accurately), we found that it speeds up the heuristic without compromising performance.

Similar to Cordeau et al. (2001), if the next solution was formed by relocating customer $i \in N$ within or from truck $v \in V$, its relocation is forbidden for the next θ iterations by assigning a tabu status to the pair (i, v), which is included in the tabu list. Nevertheless, an aspiration criterion permits overriding a move's tabu status whenever it provides a new best-so-far solution. Notice that the use of the tabu list not only allows escaping local optima, but also reduces the possibility of solving the same LSP repeatedly.

4.4.4. Diversification

In addition to the adaptive mechanism for computing the Threshold (see Section 4.3.2), following Schneider et al. (2017), we include another diversification strategy that restarts the search from the best-so-far solution ($\mathcal{X}^*$) every time the latter has not improved during the last $\eta_{\max}$ iterations. As Schneider et al. (2017) argued, the search will likely take a different path than before, as it restarts with different tabu list entries.

4.5. Matheuristic for the truck-only version of the problem

As mentioned before, the matheuristic approach also works for the truck-only version of the problem. The difference lies in the LSP, which now defines the optimal depot return operations for a given VRP solution.

In principle, the MILP formulation of the corresponding LSP can be derived quickly by setting $\boldsymbol{u} = \boldsymbol{0}$ in (51). However, in our computational implementation of the matheuristic, we opted not to declare any drone-related variables and constraints to avoid reading variables and constraints that the preprocessor of the MILP solver will eventually remove. The resulting MILP formulation of the LSP for the truck-only version of the problem is given as follows.

$$z(\mathcal{X}) = \min_{\substack{\hat{\boldsymbol{y}}, \boldsymbol{\gamma}, \boldsymbol{\vartheta}, \\ \boldsymbol{\tau}, \boldsymbol{T}, \boldsymbol{\mathcal{T}}}} \{\mathcal{T} : (34), (41)\text{–}(43), (48), (53), (54), (59), (60), (63)\text{–}(65), (67), (69)\text{–}(72)\} \tag{68}$$

The set of Constraints (69)–(72) is defined as follows.

Modification of Constraints (52):

$$\sum_{i:(i,j)\in\hat{Y}(\mathcal{X})} \hat{y}_{ij} + \gamma_j = 1 \qquad \forall j \in N \tag{69}$$

Modification of timing Constraints (55)–(58):

$$T_j \geq T_i + t_{ij} + s_j + \Delta_{ij} \cdot \vartheta_j \qquad \forall i \in N, j \in N \backslash \{i\} : \exists v \in V, \tilde{x}^v_{ij} = 1 \tag{70}$$

$$T_j \geq \tau^v_0 + t_{0j} + s_j \qquad \forall v \in V, j \in N : \tilde{x}^v_{0j} = 1 \tag{71}$$

Modification of timing Constraints (61) and (62):

$$T_i \geq (r_j + t_{0i} + s_i) \cdot \hat{y}_{ij} \qquad \forall (i,j) \in \hat{Y}(\mathcal{X}) \tag{72}$$

Note that the MILP formulation (68) does not contain any big-M constraints. Also note that by redefining $a_i := r_i + t_{0i} + s_i$ (see Section 4.3.1), the lower bounds $\ell_i(\mathcal{X}_v)$ ($\forall i \in N, v \in V$) are further tightened yet the definition of the lower bound LB($\mathcal{X}$) remains the same.

5. Numerical experiments

This section presents the results of comprehensive computational experiments using the MILP formulation and the matheuristic approach for the VRPRD-DR. The MILP model and the matheuristic were programmed in Java on a workstation Intel(R) Xeon(R) Gold 5118 CPU @2.30 GHz (12 cores) with 64 GB RAM, using CPLEX 12.8, when necessary, as a MILP solver.

5.1. Instance generation and parameter definition

To evaluate the performance of the MILP model and the matheuristic, we developed instances from 10 to 100 customers based on Dayarian et al. (2020), Ulmer et al. (2019), and Archetti et al. (2018). We consider a distribution area of 20×20 km^2, where customers are distributed uniformly at random. This area size is similar to Montgomery, AL (413 km^2) and Valparaíso, Chile (402 km^2). Similar to Dayarian et al. (2020), we locate the depot outside $(-10, 10)$ the area, reflecting the common situation of last-mile distribution facilities located on

the outskirts of cities. Nevertheless, to fully assess the matheuristic compared to the MILP model on small instances, we also consider the depot at the center $(10, 10)$ and at the border $(0, 10)$ of the area.

To generate the order release dates, we follow the procedure of Archetti et al. (2018). These authors first compute the optimal TSP value (z_{TSP}) of visiting all customers in a single tour (without release dates), and then generate uniformly random integer release dates in the time interval $[0, \beta \cdot z_{\text{TSP}}]$, where β is a factor that defines the width of the interval. In our case, we obtain z_{TSP} when the depot located at the center of the area, with a constant truck travel speed of 30 km/h and Manhattan distances. Similar to Archetti et al. (2020), we create instances with $\beta \in \{0.5, 1.0, 1.5\}$.

Regarding the drone parameters, we consider the Avidrone 210TL, manufactured by Avidrone Aerospace (Avidrone Aerospace, 2021), as an example of the type of drone that could be used for resupply. This drone has a payload capacity of 25 kilograms, a flight endurance of 90 min, and a maximum flight speed of 100 km/h. Considering that most parcels weigh less than 2.3 kilograms (Poikonen & Golden, 2020), this leads to a maximum drone load capacity (Q) of about 10 orders. However, given that we evaluate the MILP model performance over small instances of up to 20 customers, we experiment with $Q = 2$ orders for such instances. We then experiment with $Q = 10$ orders when solving larger instances using the matheuristic. We assume that all orders are interchangeable according to weight and physical size (i.e., $q_i = 1, \forall i \in N$). Furthermore, as commonly used in drone routing research (Macrina et al., 2020), we employ a constant drone speed of 60 km/h, using the Euclidean metric to calculate the drone distances. This should be an achievable average speed for the Avidrone 210TL. The time needed to receive a drone en route or at the depot is fixed at $\delta_{\text{m}} = \delta_{\text{d}} = 5$ min.

Based on Moshref-Javadi et al. (2020), we consider a fixed customer service time of three minutes (i.e., $s_i = 3, \forall i \in N$). To enable a fair comparison with the truck-only scenario, the time to receive a truck at the depot is fixed at $\delta_{\text{t}} = 5$ min, meaning that the time for receiving a drone en route (δ_{m}) or at the depot (δ_{d}) is equal to the time for receiving a truck at the depot (δ_{t}). Finally, we set a big-M value as $M = \max_{i \in N}\{r_i\} + z_{\text{TSP}}$ in all MILP models. Here, M represents the completion time if all orders to be released are waited for at the depot and a single truck then conducts an optimal TSP tour.

5.2. Effectiveness of the matheuristic

In this section, we compare the results of the MILP model and the matheuristic over instances of up to 20 customers (a detailed repository of the test instances and results can be found at GitHub at https://github.com/jcpinap/VRPRD-DR). The effectiveness of valid inequalities (44)–(49) is reported in Table V.7 of Online Appendix A, which shows that adding these constraints substantially improves the computational performance of the MILP model.

Regarding the choice of parameters and termination criteria of the matheuristic, we consider:

- Initial VRP solution (Section 4.4.1): we used the insertion heuristic I1 of Solomon (1987). After preliminary experiments, we set the parameters $\mu = \lambda = 1$, $\alpha_1 = 0.25$, and $\alpha_2 = 0.75$ (see Solomon, 1987 for details).
- Tabu tenure (Section 4.4.3): similar to Cordeau et al. (2001), we set the tabu tenure as $\theta = [7.5 \cdot \log_{10}(n)]$ iterations, where n is the number of customers and $[\cdot] : \mathbb{R} \to \mathbb{Z}$ is a function that maps each real to its nearest integer.
- Diversification mechanisms: to compute the threshold (see Section 4.3.1) and restart the search (see Section 4.4.4), we set $\eta_{\max} = 100$ iterations.
- Cutoff strategy (Section 4.3.2): we set $\varepsilon = 0.5$ min to abort the execution of the LSP as soon as CPLEX corroborates that no node of the branch-and-bound tree meets the best-so-far solution's objective value minus 30 s.
- Termination criteria: we stop the matheuristic after $m(n)$ min or when the completion time of the best-so-far solution equals the lower bound set by Proposition 1, whichever comes first. To allow a fair comparison with the MILP model and for uniformity, we set $m(10) = 2.5$ min, corresponding to the maximum CPU time required by CPLEX to solve an instance of 10 customers to optimality using the MILP model. Additionally, we set $m(15)$ and $m(20)$ to 60 min, which was the maximum run-time allowed for CPLEX to solve the MILP model.

5.2.1. Comparing solutions of 10-customer instances

We begin by reporting the performance of the matheuristic compared with the optimal results of the MILP models from the truck-and-drone and the truck-only versions of the problem (we refer the reader to Online Appendix B for a detailed description of the truck-only MILP model). Considering two delivery trucks and two resupply drones with a load capacity of two orders each, each row of Table 3 reports the average results of 15 instances, resulting from five instances per depot location (center, border, and outside). The column "Run time" reports the average CPU time required by CPLEX to solve the corresponding MILP model to optimality. We note that there is no significant difference in run times and optimality gaps between the three depot locations. Additionally, the average computational time (in seconds) of the matheuristic to find the best overall solution is reported in the "TB (s)" column. The column "Gap (%)" reports the average optimality gap and the column "Opt (%)" the percentage of optimal solutions found by the matheuristic.

For the truck-and-drone version of the problem, Table 3 shows that the matheuristic solved 93% of the instances to optimality (42 out of 45), with only a trivial gap. Furthermore, it finds the best solution in 8 s on average. On the other hand, the matheuristic solved all instances to optimality for the truck-only version. Nonetheless, it requires (on average) a few seconds more to find optimal solutions when $\beta = 1.0$.

5.2.2. Comparing solutions of 15- and 20-customer instances

Now, we present the comparison for instances of 15 and 20 customers, considering a maximum run-time of one hour for both the matheuristic and the MILP models. For instances of 15 customers, CPLEX found 16 (out of 45) and 11 (out of 45) optimal solutions using the truck-and-drone and the truck-only MILP models, respectively. However, for both versions of the problem, CPLEX failed to find any optimal solutions for instances of 20 customers, showing that the MILP models quickly become intractable as the number of customers increases (see our aforementioned GitHub repository for detailed results).

Table 4 reports the computational performance of the MILP model and the matheuristic for each version of the problem. Columns "Min", "Mean", and "Max" report the minimum, mean, and maximum relative percentage differences between the objective value reached by the matheuristic and the objective value reached by the corresponding MILP model in one hour (a negative value means that the matheuristic outperforms the MILP model). For instances of 15 customers, the matheuristic obtained similar results to the MILP models, with better performance as the value of β decreases. This is likely because the performance of the MILP model decreases as β decreases. Furthermore, the matheuristic found 14 of 16 known optimal solutions (with an average optimality gap of only 0.4% for the other two instances) for the truck-and-drone scenario, and 10 of 11 known optimal solutions for the truck-only scenario (with an optimality gap of 1.4% for the non-optimally solved instance). For instances of 20 customers, the matheuristic consistently outperforms the MILP models. In the truck-and-drone version of the problem, the matheuristic improves the MILP model results by 5.0% on average, while improving 1.0% on average for the truck-only version. Finally, regarding computational times, the matheuristic found the best overall solutions in 6.5 and 11.3 min on average for instances of 15 customers for the truck-and-drone and

Table 3
Matheuristic performance for instances of 10 customers for each version of the problem, considering two trucks and two drones with a load capacity of two orders each, for different spreads of release dates (β).

β	Truck-and-drone				Truck-only			
	MILP model (1)–(49)	Matheuristic			MILP model	Matheuristic		
	Run time (s)	TB (s)	Gap (%)	Opt (%)	Run time (s)	TB (s)	Gap (%)	Opt (%)
0.5	23.2	10.7	0.00	100.0	7.7	7.6	0.00	100.0
1.0	25.2	11.3	0.06	86.7	6.5	19.0	0.00	100.0
1.5	3.8	1.4	0.02	93.3	4.2	0.6	0.00	100.0
Average	**17.4**	**7.8**	**0.03**	**93.3**	**6.1**	**9.1**	**0.00**	**100.0**

Table 4
Computational performance of the MILP models and the matheuristic over 15 instances of 15 and 20 customers, considering two trucks and two drones with a load capacity of two orders each.

n	β	Truck-and-drone						Truck-only					
		MILP model (1)–(49)		Matheuristic				MILP model		Matheuristic			
		Run time (s)	Gap (%)	TB (s)	Min	Mean	Max	Run time (s)	Gap (%)	TB (s)	Min	Mean	Max
15	0.5	3600.4	16.3	505.8	−6.04	−1.95	1.21	3607.1	12.0	959.2	−3.43	−0.47	1.18
	1.0	2677.2	5.6	621.2	−6.32	−0.82	0.55	3126.7	5.6	740.7	−2.44	0.24	4.72
	1.5	878.9	0.4	46.1	0.00	0.02	0.31	1453.3	1.0	342.8	−0.62	0.07	1.40
20	0.5	3604.1	26.0	906.6	−12.12	−5.90	1.57	3611.3	17.3	888.6	−4.17	−1.68	1.89
	1.0	3604.2	13.6	744.9	−10.77	−6.23	1.09	3610.8	7.9	894.2	−6.75	−0.91	4.44
	1.5	3603.1	6.1	560.3	−9.97	−3.12	0.00	3613.6	4.6	509.8	−2.12	−0.38	0.62

truck-only scenarios, respectively. For instances of 20 customers, the best overall solution was found in 12.3 and 12.7 min on average for these two scenarios. We refer the reader to Online Appendix D for further information on the relative percentage difference (over time) between the current best solution and the best overall solution found by the matheuristic. Results show that, for instances of up to 100 customers, the solutions found by the matheuristic after 7 min are (on average) within 5% of the best solutions found after one hour. This suggests that, in practice, managers could run the matheuristic for just several minutes and obtain high-quality solutions. These computational times are modest and the matheuristic's flexibility in this regard is an advantage that allows high-quality solutions to be achieved within quite constrained timeframes.

In summary, the approaches perform very similar for 10- and 15-customer instances but quite different when the size reaches 20 customers (and we can surmise beyond). As realistically sized instances would be 20 or above, it is imperative to have an effective heuristic, such as this one.

5.3. Large instances and managerial insights

Given the encouraging results of the matheuristic for small instances, we investigate instances of up to 100 customers. For these, we set the depot outside of the distribution area and use a drone load capacity of 10 orders, which is consistent for drones such as the Avidrone 210TL (Avidrone Aerospace, 2021). To make a fair comparison, we used 60 min as the termination criteria for both the truck-and-drone and the truck-only versions of the matheuristic (see our GitHub repository for detailed results of the matheuristic).

We begin by investigating the effects of the spread of order release dates over instances of up to 50 customers. To this end, we consider two delivery trucks and two resupply drones with a load capacity of 10 orders each. Fig. 5 shows the distribution of the percentage of completion time savings compared with the truck-only delivery system, where the trucks must return to the depot to collect any newly released orders. Results indicate that drone resupply reduces completion times for all instances tested (90 in total), regardless of the value of the parameter β. The maximum saving percent is 11.0% (equivalent to 46 min) and occurs for an instance of 30 customers with $\beta = 1.0$.

For instances of 40 and 50 customers, Fig. 5 also shows that the median percent time saved is greater when $\beta = 1.0$. Intuitively, if $\beta = 0$ (i.e., all orders are available for dispatch at the beginning of the working day), there is no need for drone resupply or even truck returns to the depot; therefore, savings are zero. Conversely, if β is large enough, the completion time of the delivery process will be determined almost entirely by the release date of the last orders, making the benefits of drone resupply marginal or almost none. Thus, the savings derived from drone resupply are likely concave with respect to β, as shown empirically in Fig. 5.

To understand the delivery process in more detail, Table 5 shows the average fleet utilization for instances of 50 customers (similar conclusions were obtained for instances of 30 and 40 customers). Although the number of truck returns to the depot is drastically reduced compared with the truck-only delivery system, trucks still need to return to the depot when drone resupply is considered (when appeared, returns to the depot were performed by any truck and not only by the one that returned to the depot last). This occurs mainly because delivery drivers, at each resupply operation, potentially need to wait for the drones to arrive at the rendezvous location, which increases the completion time of their master tours. Therefore, returning to the depot can be effective in reducing (potentially excessive) truck waiting times for drones.

Table 5 also shows that trucks return to the depot more frequently as the order release dates are more spread out over time. However, the average number of orders collected per return substantially decreases as β increases, and the same occurs with drones in terms of the number of orders resupplied per sortie. These results indicate that consolidation (i.e., waiting for additional orders before dispatching the vehicles) becomes less relevant as release dates are more dispersed over time, which is consistent with the conclusions that Archetti et al. (2018) obtained for the truck-only (and single-truck) version of the problem.

Now, to analyze the behavior of drone resupply over large instances of up to 100 customers, we consider having an unlimited number of drones (with a load capacity of 10 orders each) available at the depot (see Online Appendix C for a detailed description of the LSP model for this case). To gain insights regarding the necessary number of drones to fully replace truck return trips to the depot, we are not considering depot returns in the unlimited drone fleet scenario; in all other aspects, we maintained our previous modeling considerations (see Section 3).

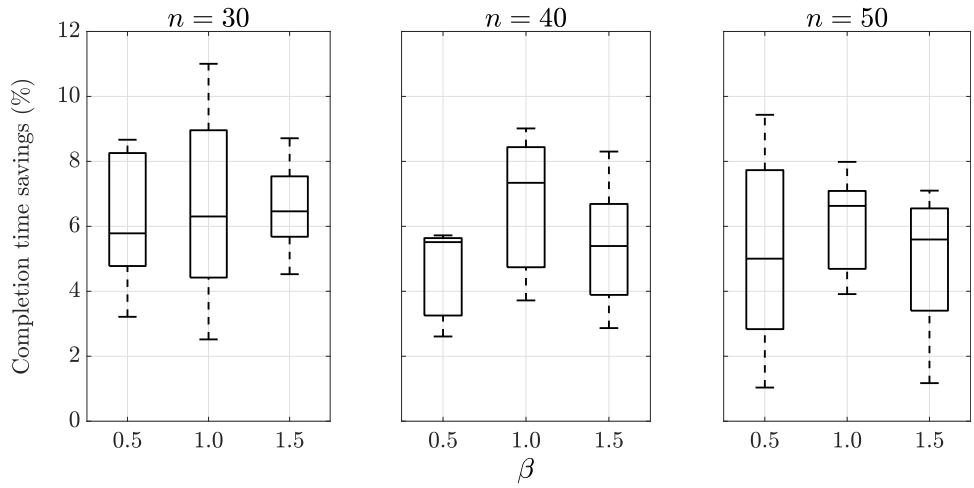

Fig. 5. Distribution of the completion times savings (%) over 10 instances for each number of customers (n), considering two trucks and two drones with a load capacity of 10 orders each. The depot is located outside of the distribution area.

Table 5
Average fleet utilization for 10 instances of 50 customers, considering two trucks and two drones with a load capacity of 10 orders each, with the depot located outside of the distribution area.

β	Truck-and-drone				Truck-only	
	Total drone sorties	Orders sent per sortie	Total truck returns	Orders collected per return	Total truck returns	Orders collected per return
0.5	6.7	6.8	0.4	5.8	6.7	6.5
1.0	9.4	4.4	2.3	2.7	9.7	4.7
1.5	12.7	3.3	2.8	1.7	11.1	3.7

Table 6
Average results over 10 instances of 50, 75, and 100 customers, considering an unlimited number of drones and $\beta = 1.0$. The depot is located outside of the distribution area.

n	Truck-and-drone						Truck-only			
	Orders loaded (%)		Fleet utilization			Improvement	Orders loaded (%)		Fleet utilization	
	Before leaving the depot	Resupplied by drones	Drones required	Total drone sorties	Orders sent per sortie	Savings (%)	Before leaving the depot	Via depot return	Total truck returns	Orders collected per return
50	3.6	96.4	3.2	12.9	3.9	8.1	11.0	89.0	9.7	4.7
75	3.2	96.8	3.3	14.0	5.3	5.7	9.2	90.8	11.6	5.9
100	2.3	97.7	3.4	15.2	6.4	4.3	8.4	91.6	12.4	7.5

Table 6 reports the average results over 10 instances of 50, 75, and 100 customers. For both the truck-and-drone and the truck-only scenarios, the percentage of orders (with respect to the total number of orders) loaded before the trucks first depart the depot decreases as the number of customers increases. Thus, along with our previous findings from Table 5, we conclude that consolidation loses relevance as release dates are more dispersed over time and more customers require service. These conclusions are both intuitive. Furthermore, the percentage of orders resupplied by drones in the truck-and-drone model is similar to that collected by trucks via depot returns in the truck-only model, indicating that drone resupply is preferable to instructing trucks to return to the depot to collect newly available orders.

Table 6 also reports the actual minimum number of drones required, which we derived through a post-processing procedure over the final solutions obtained by the matheuristic. These results show that last-mile operators need a relatively small fleet of drones (two to four in our studies) to fully replace truck return trips to the depot. For instances of 100 customers, these drones performed a total of 15.2 sorties, sending 6.4 orders per sortie on average (which is feasible for drones such as the Avidrone 210 TL). To put this in perspective, the capital cost of a drone is relatively modest compared to other modes of transport and is a small fraction of what a truck would cost.

For the truck-only scenario, trucks performed a total of 12.4 return trips on average during the working day (roughly six returns per truck), collecting 7.5 orders per return on average. Overall, the average percentage of completion time savings using drones diminishes as the size of the instance increases. The increase in the number of customers within the same distribution area (20×20 km^2) increases customer density, making drones relatively less advantageous. However, the benefit of drones goes beyond delivery time savings as there are positive impacts on sustainability measures such as traffic and emissions.

6. Conclusions

In this paper, we investigated a truck-and-drone delivery system in which trucks perform last-mile deliveries while drones send newly released orders to those trucks. These orders could represent make-to-order items that have just finished being prepared, or items that just arrived from the previous leg of the supply chain. This approach leverages the benefits of combining trucks and drones for last-mile logistics and, most importantly, addresses the unresolved issue of how customers could safely receive orders from drones.

Considering a fleet of trucks and drones, we formally introduced the Vehicle Routing Problem with Release Dates and Drone Resupply (VRPRD-DR). This problem consists of routing a fleet of trucks to deliver orders that become available throughout the working day. To manage the arrival of new orders, the VRPRD-DR considers that en

route trucks can return to the depot to collect them, or these orders can be resupplied to these trucks along their routes via drones dispatched from the depot. Assuming that order release dates are known at the planning time, we developed a MILP formulation to minimize the completion time of the delivery process, along with several valid inequalities to strengthen its computational performance. Numerical experiments showed that the MILP formulation can obtain optimal solutions in a few seconds for instances of 10 customers. Nevertheless, the MILP becomes intractable when solving instances of 20 customers or more, sizes of practical interest.

To address larger instances, we developed a unified matheuristic approach that provides high-quality solutions for both the truck-and-drone and the truck-only versions of the VRPRD-DR. In this approach, we iteratively (and strategically) modify the truck routes and define optimal loading operations (truck depot returns and drone resupplies) through a fast subordinate MILP model. For instances of 10 customers, computational experiments showed that the matheuristic solved most to optimality. As expected, the matheuristic outperformed the MILP formulation when solving instances of 15 and 20 customers.

Investigations with instances of up to 100 customers showed that, compared to the truck-only scenario, drone resupply reduces completion times and the number of times trucks need to return to the depot. Furthermore, our results suggest that the fraction of orders resupplied via drones is similar to the fraction collected via depot returns in the truck-only scenario, implying that it is generally more desirable to collect these orders via drone resupply rather than through depot returns. Our results also show that consolidation loses relevance as order release times are more dispersed over time and more customers require service.

As our work is one of the first to investigate using drones for resupply, several future extensions to the VRPRD-DR can be readily suggested. For instance, we focus on minimizing the total duration (i.e., makespan) of the delivery process, without imposing a particular preference for drone resupply over depot return. In practice, however, drone resupply represents a more sustainable and potentially much cheaper alternative than instructing trucks to return to the depot periodically during the working day. Thus, other performance metrics can be proposed to better account for different decision-maker preferences and urban regulations. Furthermore, minimum consolidation levels could be imposed to dispatch trucks and drones, as last-mile logistics operators commonly seek high vehicle utilization. Other extensions to enrich the model include allowing drones to resupply multiple trucks per sortie (which might be worth studying especially when considering other resupply vehicle types, such as motorcycles or cargo bikes) and allowing drone resupply at locations other than customer nodes, though these extensions would involve considerable complications (see, e.g., Schermer et al., 2019a, which studies the benefits of launching and retrieving drones in arcs for an extension of the Flying Sidekick TSP). Additionally, useful future work could investigate the case where travel times are uncertain (see, e.g., Yang et al., 2023). Finally, given the rise of same-day delivery services, the stochastic and dynamic version of the VRPRD-DR is a highly relevant avenue for future research.

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Appendix A. Additional Model Formulations and Results

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